

THARANI RAVI

ASPIRING DATA ENGINEER

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with internship experience contributing to ETL workflows, data ingestion, validation, and workflow orchestration using Python, SQL, PostgreSQL, Prefect, and Docker. Experienced in developing data-driven and IoT applications, with a strong foundation in machine learning, data preprocessing, and Git-based collaborative development. Seeking to build scalable data engineering solutions.

TECHNICAL SKILLS

Programming: Python, SQL

Data Engineering: ETL, Data Pipelines, Data Ingestion, Data Transformation, Data Validation, Workflow Automation, Workflow Orchestration

Databases: PostgreSQL

Libraries: Pandas (Basic), NumPy (Basic), Plotly (Basic)

Tools: Docker, Git, GitHub, Prefect, Jupyter Notebook

Machine Learning: Data Preprocessing, Predictive Modeling

Other: REST APIs, IoT, Arduino, Data Analytics

PROFESSIONAL EXPERIENCE

AI & Data Analytics Intern — Kaar Technologies, Chennai

2026 – Present

- Contributed to migrating an existing ETL workflow into a standardized Prefect-based orchestration framework by assisting with schema mapping and transformation logic. Implemented PostgreSQL-backed data ingestion within a Dockerized development environment and supported scheduled workflow orchestration.
- Automated generation of structured technical documentation using Python, and validated output completeness against a 250-item source dataset.
- Debugged Docker-based development environments to improve pipeline execution reliability during development.

PROJECTS

Smart Traffic Management System | Python, Machine Learning, Data Preprocessing

- Pre-processed traffic datasets, engineered predictive features, and trained a machine learning model to optimize adaptive signal timing. Ran simulations to evaluate model-driven traffic control logic and validate signal timing adjustments against vehicle density.

GPS-Based Power Theft Detection and Alert System | GPS, GSM, Arduino, IoT

- Integrated GPS and GSM modules with Arduino to enable location-tagged power theft detection and real-time alert generation.
- Presented and published the system at an IEEE conference, demonstrating embedded real-time data collection and alerting logic.

PATENT

- Fall Detection Using LiDAR — Patent Applied (Application No. 202541013131A).

PUBLICATIONS

- IEEE Conference Publication — "GPS-Based Power Theft Detection and Alert System Using GSM."

ACHIEVEMENTS

- Best Paper Award — INNOVISTA'23, Robotic Surgery Advances.
- IEEE YESIST'12 Innovation Challenge — Finalist, Tunisia (2023–24).
- Smart India Hackathon — Shortlisted, College Level (2024).

EDUCATION

B.E., Electronics and Communication Engineering (2022 – 2026) — RMK Engineering College CGPA: 8.18

HSC, (2022) St. John's MHSS *Percentage: 86.33%*

SSLC,(2020) St. John's MHSS *Percentage: 90.2%*

LANGUAGES

English - Professional Proficiency

Tamil - Professional Proficiency

Hindi - Reading and Writing